

Event-Based Finite-Time Control for Nonlinear Multi-Agent Systems with Asymptotic Tracking

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Abstract—In this paper, an adaptive neural finite-time event-triggered consensus tracking problem is studied for nonlinear multi-agent systems (MASs) under directed graphs. Firstly, the unknown nonlinear functions of MASs can be approximated by neural networks. Then, a distributed adaptive event-triggered control (ETC) scheme is proposed via command filter and backstepping technique. The newly designed control scheme can not only circumvent the problem of the explosion of complexity, but also remove the singularity issue typical of conventional backstepping technique. In the meanwhile, an event-triggered mechanism with a dynamic threshold is devised to reduce the waste of network resources. Moreover, by using a novel finite-time stability criterion, it can be proved that the closed-loop system is finite-time stable and the consensus tracking errors can reach zero as time approaches to infinity. Finally, a numerical example is given to validate the feasibility of the proposed scheme.

Index Terms—Adaptive control, finite-time stability, event-triggered control, command filter, multi-agent systems.

I. INTRODUCTION

Over the last decade, the distributed consensus control of multi-agent systems (MASs) has long been a research topic attracting considerable attention [1]–[3]. As a fundamental problem in distributed control of MASs, consensus tracking (leader-follower consensus) has been extensively studied owing to its applications in flocking of social insects, cooperative control of unmanned aerial vehicles, spacecraft coordination, sensor networks and robot formation [4]–[9]. Compared with the tracking control of single system [10], the main difficulty of the consensus tracking control is to design a distributed controller on the basis of local information ensuring that all follower agents can track the desired reference trajectory (the leader). Besides, a remarkable characteristic of the aforementioned works is that the considered systems are relatively simple, where only the linear systems are taken into consideration. However, a great deal of control systems contain unknown nonlinear terms in practical engineering. Thus, the fuzzy logic system and neural network (NN) have been utilized to tackle the completely unknown nonlinear dynamics such as those in [11]–[14].

Note that all of the aforementioned works can only ensure asymptotic tracking for the controlled system. Different from the asymptotic tracking, the finite-time tracking control has plenty of merits such as higher convergence rate, superior tracking

performance, and better robustness [15]–[20]. Therefore, some finite-time control methods for MASs have been put forward in recent years. The authors of [21] investigated finite-time consensus control (FTCC) for MASs with double-integrator dynamics. In [22], a novel adaptive finite-time continuous distributed control algorithm was proposed for the MASs with parametric uncertainties. In [23], the finite-time leader-follower consensus problem was investigated for a class of high-order MASs with uncertain nonlinear dynamics. Nevertheless, there exists the problems of “explosion of complexity” and “singularity” caused by directly employing the derivatives of the virtual control signals typical of backstepping technique. To solve this problem, the command filter technique was exploited in [24], the main idea of which is to approximate the derivative of the virtual control signals through a command filter. In the meanwhile, the compensation signal is devised to decrease the error coming from the first-order filters. The distributed finite-time consensus tracking control scheme was proposed in [25] by using the command filter and backstepping method.

It should be pointed out that the designed controllers in the above results are updated in a fixed time, which results in high updating frequency of the controller and a mass of communication consumption. In many applications of MASs, each agent is equipped with limited energy and communication resources. Therefore, it is practical significance to design an adaptive controller to reduce the occupancy of network resources and improve the control efficiency. Recently, event-triggered control (ETC) strategy was introduced in [26] to tackle this problem. Based on this control approach, the control task was executed only when a prescribed triggering condition was violated, and many efforts have been made to use the advanced event-triggered sampling technology to consensus control of MASs [27]–[32]. The authors of [27] first put forward an ETC scheme to solve a single-integrator consensus problem for MASs. Then, in [28], the authors considered an average consensus problem for multiple integrators, and developed event-based control algorithms for each agent with undirected graphs. The authors of [29] proposed a new combinational measurement approach to solve the consensus problems by an event-triggered controller for MASs. The proposed control scheme was associated with a considerably low-computational burden as every agent is only triggered at their own event time. The authors of [30] creatively established a novel event-triggered framework by introducing distributed event processors, which represented a significant breakthrough in the distributed adaptive ETC for MASs with directed graph. By establishing a new definition condition of consensus control, the ETC problem

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for a class of nonlinear MASs was studied in [31]. Recently, an adaptive event-triggered protocol with a variable threshold was established for MASs in [32] by exploiting the hyperbolic tangent function and backstepping technique. In general, the obtained results in the framework of [25]–[32] are relatively conservative. Firstly, only bounded error tracking could be given in the aforementioned results, and zero-error tracking could not be achieved in [25]–[32]. Secondly, the explosion of complexity problem appears in above-mentioned control schemes [25]–[32]. Thirdly, the aforementioned control schemes can only achieve asymptotical consensus tracking, which implied that consensus tracking cannot be obtained in finite-time. Hence, how to design a new adaptive finite-time consensus tracking control scheme for MASs without the explosion of complexity problem is a challenging task that needs further investigation.

Based on the discussions above, a novel command filtered-based ETC scheme is first proposed to achieve a fast convergence rate. The novelty of this study against the existing works can be summarized as follows:

(1) A novel command filter-based adaptive neural finite-time distributed controller is constructed for MASs to ensure that all signals of the closed-loop system are bounded in finite-time. In contrast to the ultimately bounded control results established in [21]–[23], [25], the proposed distributed control algorithm guarantees the consensus tracking errors can reach zero as time approaches to infinity. This is achieved by using a modified finite-time lemma and some smooth functions introduced in the controller.

(2) Compared with the existing finite-time adaptive control schemes [22], [23], the “singularity” and “explosion of complexity” problems in backstepping are successfully circumvented by using the command filter technique. In addition, a novel finite-time error compensation mechanism is proposed to reduce the influence of filtering errors. Different from the command filtering technique [25], our designed finite-time compensation mechanism can effectively decrease the chattering phenomenon, this is done by introducing a roust term. Besides, the compensated error can achieve the asymptotical convergence.

(3) In contrast to the ETC scheme [32] where the controller is dependent on the residual term of event-triggered mechanism, no information on the residual term of event-triggered mechanism, are required by the proposed event-triggered controller in this paper. Therefore, it provides a different insight on how to deal with ETC control in an adaptive manner for a more general class of nonlinear MASs.

II. PROBLEM STATEMENT

A. System Dynamics

This paper studies the consensus tracking control problem for a class of nonlinear MASs. Suppose that the nonlinear MASs are composed of one leader (labeled by $r(t)$) and N followers (labeled from 1 to N). The dynamics of the i th follower is modeled by the following differential equation:

$$\begin{aligned} \dot{x}_{i,q} &= x_{i,q+1} + f_{i,q}(\bar{x}_{i,q}), \quad q = 1, 2, \dots, n_i - 1 \\ \dot{x}_{i,n_i} &= u_i + f_{i,n_i}(x_i) \\ y_i &= x_{i,1}, \quad i = 1, 2, \dots, N \end{aligned} \quad (1)$$

where $\bar{x}_{i,q} = [x_{i,1}, \dots, x_{i,q}]^T \in R^q$; $x_i = [x_{i,1}, \dots, x_{i,n_i}]^T \in R^{n_i}$ denotes the state vector; $u_i \in R$ and $y_i \in R$ are the control input and system output, respectively. $f_{i,q}(\bar{x}_{i,q}) \in R$ is an unknown continuous nonlinear function. Besides, the leader's dynamics is represented as $r(t) \in R$.

The purpose of this paper is to devise an event-triggered distributed controller such that follower y_i can track the leader $r(t)$ in finite-time, and the tracking errors $\delta_i = y_i - r(t)$ can reach zero as time approaches to infinity. At the same time, all signals in the resulting system are bounded and the Zeno behavior is excluded.

Based on the above purpose, some useful assumptions and lemmas are presented below.

Assumption 1: There exists an unknown positive constant M_1 such that $|r(t)| \leq M_1$.

Remark 1: Assumption 1 indicates that the reference signal $r(t)$ is bounded and continuous. Such a condition is much weaker than that of [34]–[36] where the reference signals are linearly parameterized. Moreover, Assumption 1 is particularly important for the following stability and controller design.

Lemma 1: [19] For any $\tilde{a} = a - \hat{a}$, and $0 < h = 2p - 1$ with $\frac{1}{2} < p < 1$, the following relation holds

$$\tilde{a}(a - \tilde{a})^h \leq c_0 a^{2p} - d_0 \tilde{a}^{2p} \quad (2)$$

where $c_0 = \frac{1}{1+h} - \frac{2^{h-1}}{1+h} + \frac{2^{2h-h^2-1}}{(1+h)^2} + \frac{h}{(1+h)^2}$ and $d_0 = \frac{(2^{h-1}-2^{(1+h)(h-1)})}{1+h} > 0$.

Lemma 2: [33] If $f(t)$ is a continuous function on $[0, +\infty)$ and the inequality $\lim_{t \rightarrow +\infty} \int_0^t f(t)dt < \bar{f} < \infty$ holds with $\bar{f}$ a constant. Then, it holds that $\lim_{t \rightarrow +\infty} f(t) = 0$.

Lemma 3: Considering the nonlinear system $\dot{x} = \phi(x)$, there exists a continuous nonlinear function $V(x(t)) : R^n \rightarrow R^+ \cup \{0\}$ and several real numbers $a_0 > 0$, $\frac{1}{2} < p < 1$ and $b_0 > 0$ such that $\dot{V}(x(t)) \leq e^{-\lambda t}(a_0 V^p(x(t)) + b_0) = a V^p(x(t)) + b$, with $a = a_0 e^{-\lambda t}$, $b = b_0 e^{-\lambda t}$. Then, $V(x(t))^p \leq \frac{b_0}{(1-\varrho)a_0}$ in finite-time. The convergence time can be defined as

$$T = \frac{1}{\lambda} \ln \frac{\varrho a_0}{\frac{\lambda}{1-p}((\frac{b_0}{(1-\varrho)a_0})^{\frac{1-p}{p}} - V^{1-p}(0)) + \varrho a_0} \quad (3)$$

where $0 < \varrho < 1$.

Proof: According to $\dot{V}(x(t)) \leq e^{-\lambda t}(a_0 V^p(x(t)) + b_0)$, we can obtain that

$$e^{\lambda t} \dot{V}(x(t)) \leq a_0 V^p(x) + b_0 \quad (4)$$

Then, define two compact sets $\Psi_1 = \{t | V^p(x(t)) \leq \frac{b_0}{(1-\varrho)a_0}\}$ and $\Psi_2 = \{t | V^p(x(t)) \geq \frac{b_0}{(1-\varrho)a_0}\}$.

(1) If $t \in \Psi_2$, we have

$$\begin{aligned} e^{\lambda t} \dot{V}(x(t)) &\leq -\varrho a_0 V^p(x(t)) - (1-\varrho)a_0 V^p(x(t)) + b_0 \\ &\leq -\varrho a_0 V^p(x(t)) \end{aligned} \quad (5)$$

which is equivalent to

$$\dot{V}(x(t)) V^{-p}(x(t)) \leq -e^{-\lambda t} \varrho a_0 \quad (6)$$

By integrating the left and right sides of (6) on $[0, T)$, one has

$$\int_0^T \dot{V}(x(t)) V^{-p}(x(t)) dt \leq - \int_0^T \varrho a_0 e^{-\lambda t} dt \quad (7)$$

Further a simple derivation shows

$$\frac{1}{1-p}(V^{1-p}(x(T)) - V^{1-p}(x(0))) \leq \varrho a_0(e^{-\lambda T} - 1) \quad (8)$$

The above inequality is satisfied if the time T is selected as

$$T \leq \frac{1}{\lambda} \ln \frac{\varrho a_0}{\frac{\lambda}{1-p}(\frac{b_0}{(1-\varrho)^a} \frac{1-p}{p} - V^{1-p}(0)) + \varrho a_0} \quad (9)$$

That is, when $t \geq T$, we have $V^p(x(t)) \leq \frac{b_0}{(1-\varrho)a_0}$.

(2) If $t \in \Psi_1$, by using LaSalle's invariance principle, the trajectory of $V(x(t))$ does not overstep the set Ψ_1 .

From the above analysis, we can get $V^p(x(t)) \leq \frac{b_0}{(1-\varrho)a_0}$ holds for $\forall t \geq T$. Therefore, the nonlinear system $\dot{x} = \phi(x)$ is finite-time stable and the convergence time T is defined by (3).

Remark 2: Lemma 3 can be regarded as a generalization of the traditional finite-time stability criterion i.e. $\dot{V}(x) \leq -aV^p(x) + b$ where a and b are known constants. In this paper, a and b can be allowed to be time-varying functions, which plays a key role in the subsequent stability analysis.

Lemma 4: [32] Consider an unknown continuous function $F_{i,q}(X_{i,q})$ defined over a compact set Ω , there exists the following equation such that for $\forall \varepsilon_{i,q} > 0$

$$F_{i,q}(X_{i,q}) = W_{i,q}^T \psi_{i,q}(X_{i,q}) + \delta_{i,q}(X_{i,q}), |\delta_{i,q}(X_{i,q})| \leq \varepsilon_{i,q} \quad (10)$$

where $W_{i,q} \in R^M$ is the ideal weight vector with M being the number of nodes, $\delta_{i,q}(X_{i,q})$ represents the approximation error, and $\psi_{i,q}(X_{i,q}) \in R^M$ denotes the basis function vector.

B. Graph theory

The interconnection topology among the N agents can be represented by a digraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with the node set $\mathcal{V} = 1, \dots, N$ and the edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. The adjacency matrix $A = (a_{ij})$ associated with $\mathcal{G}$ is defined as $a_{ij} > 0$, if $(j, i) \in \mathcal{E}$; otherwise, $a_{ij} = 0$. Moreover, we presume $a_{ii} = 0$. The Laplacian matrix of $\mathcal{G}$ is given as $\mathcal{L} = \mathcal{D} - \mathcal{A}$, where the degree matrix $\mathcal{D} = \text{diag}\{d_1, d_2, \dots, d_N\}$ with $d_i = \sum_{j=1}^N a_{ij}$. The communication topology of the leader and followers can be modeled by an extension directed graph $\bar{\mathcal{G}} = (\bar{\mathcal{V}}, \bar{\mathcal{E}})$, where $\bar{\mathcal{V}} = \{0, 1, \dots, N\}$. The Laplacian matrix of $\bar{\mathcal{G}}$ is designed as $\bar{\mathcal{L}} = \begin{bmatrix} 0 & 0_{1 \times N} \\ -b & \mathcal{L} + \mathcal{B} \end{bmatrix}$, in which $b = [b_1, b_2, \dots, b_N]^T$, $\mathcal{B} = \text{diag}\{b_1, b_2, \dots, b_N\}$ with $b_i = 1$ if follower i can receive information from the leader, otherwise, $b_i = 0$.

Assumption 2: The augmented graph $\bar{\mathcal{G}}$ contains a spanning tree with the root node being the leader node 0.

C. Event-Triggered Protocols

Assume that the controller is implemented on a digital platform. The event-triggered error of system (1) is defined as:

$$e_{is} = u_{is} - u_i(t), t \in [t_{i,k}, t_{i,k+1}) \quad (11)$$

where $u_{is} = u_i(t_{i,k})$ is the sampled control, $t_{i,k}, t_{i,k+1}$ are the triggering time and $u_i(t)$ is the continuous control of agent i . A triggering interval $[t_{i,k}, t_{i,k+1})$ can be thought of as a cycle,

where the error signal e_{is} is varied randomly and reset to 0 at each $t_{i,k}$. In this paper, the ETC protocol is designed as

$$u_{is} = u_i(t_{i,k}), \quad \forall t \in [t_{i,k}, t_{i,k+1}) \quad (12)$$

$$t_{i,k+1} = \inf \{t \in \mathbb{R} | |e_{is}(t)| \geq \delta_{i,1} |u_i(t)| + \delta_{i,2} e^{-\beta t}\} \quad (13)$$

where $0 \leq \delta_{i,1} < 1$, $\delta_{i,2} > 0$ and $\beta > 0$ are design parameters.

III. DISTRIBUTED CONTROL PROTOCOL

In this section, a command filter-based distributed finite-time ETC scheme is presented with the aid of backstepping algorithm. At first, the consensus tracking errors for the i th follower are described by:

$$\begin{aligned} z_{i,1} &= \sum_{j=1}^N a_{ij} (y_i - y_j) + b_i (y_i - r) \\ z_{i,q} &= x_{i,q} - \bar{\alpha}_{i,q}, \quad q = 2, \dots, n_i - 1 \\ z_{i,n_i} &= x_{i,n_i} - \bar{\alpha}_{i,n_i} \end{aligned} \quad (14)$$

where $i = 1, 2, \dots, N$, $\bar{\alpha}_{i,q}$ is the output of the following first-order command filter

$$\epsilon_{i,q} \dot{\bar{\alpha}}_{i,q} + \bar{\alpha}_{i,q} = \alpha_{i,q-1}, \bar{\alpha}_{i,q}(0) = \alpha_{i,q-1}(0) \quad (15)$$

where $\epsilon_{i,q} > 0$ is a design constant, and $\alpha_{i,q}$ is the virtual controller. The virtual controller $\alpha_{i,q}$ and the adaptive laws $\hat{\theta}_{i,q}$ and $\hat{\rho}_{i,q}$ are designed in the following forms:

$$\alpha_{i,1} = \frac{1}{(b_i + d_i)} \left(-k_{i,1} z_{i,1} - \lambda_{i,1} \sigma_{i,1} \eta_{i,1}^{2p-1} - l_{i,1} \frac{(1 + \sigma_{i,1}) s_{i,1}^2}{\sqrt{s_{i,1}^2 + \sigma_{i,1}^2}} - \frac{\eta_{i,1} v_{i,1}^2}{\sqrt{\eta_{i,1}^2 v_{i,1}^2 + \sigma_{i,1}^2}} \right) \quad (16)$$

$$\alpha_{i,q} = -k_{i,q} z_{i,q} - c_{i,q} z_{i,q-1} - \lambda_{i,q} \sigma_{i,q} \eta_{i,q}^{2p-1} - l_{i,q} \frac{(1 + \sigma_{i,q}) s_{i,q}^2}{\sqrt{s_{i,q}^2 + \sigma_{i,q}^2}} - \frac{\eta_{i,q} v_{i,q}^2}{\sqrt{\eta_{i,q}^2 v_{i,q}^2 + \sigma_{i,q}^2}} \quad (17)$$

$$\begin{aligned} \alpha_{i,n_i} &= z_{i,n_i-1} + k_{i,n_i} z_{i,n_i} + \lambda_{i,n_i} \sigma_{i,n_i} \eta_{i,n_i}^{2p-1} \\ &+ \frac{\eta_{i,n_i} v_{i,n_i}^2}{\sqrt{\eta_{i,n_i}^2 v_{i,n_i}^2 + \sigma_{i,n_i}^2}} + l_{i,n_i} \frac{\sigma_{i,n_i} s_{i,n_i}^2}{\sqrt{s_{i,n_i}^2 + \sigma_{i,n_i}^2}} \end{aligned} \quad (18)$$

$$\dot{\hat{\theta}}_{i,q} = \gamma_{i,q} \xi_{i,q} \eta_{i,q} - \sigma_{i,q} \hat{\theta}_{i,q}^{2p-1} \quad (19)$$

$$\dot{\hat{\rho}}_{i,q} = \frac{r_{i,q} \eta_{i,q}^2}{\sqrt{\eta_{i,q}^2 + \sigma_{i,q}^2}} - \sigma_{i,q} \hat{\rho}_{i,q}^{2p-1} \quad (20)$$

where $k_{i,q}$, $\lambda_{i,q}$, $\gamma_{i,q}$ and $r_{i,q}$ are positive constants. $\xi_{i,q} = \frac{\eta_{i,q} \psi_{i,q}^T \psi_{i,q}}{\sqrt{\eta_{i,q}^2 \psi_{i,q}^T \psi_{i,q} + \sigma_{i,q}^2}}$. $\sigma_{i,q}$ are chosen as negative exponential function $e^{-\lambda t}$ with $\lambda > 0$. And $v_{i,1}$, $v_{i,q}$ and v_{i,n_i} represent the intermediate variables, the specific definition is given below

$$v_{i,1} = \hat{\theta}_{i,1} \xi_{i,1} + \frac{\hat{\rho}_{i,1} \eta_{i,1}}{\sqrt{\eta_{i,1}^2 + \sigma_{i,1}^2}} \quad (21)$$

$$v_{i,q} = \hat{\theta}_{i,q} \xi_{i,q} + \frac{\hat{\rho}_{i,q} \eta_{i,q}}{\sqrt{\eta_{i,q}^2 + \sigma_{i,q}^2}} \quad (22)$$

$$v_{i,n_i} = \hat{\theta}_{i,n_i} \xi_{i,n_i} + \frac{\hat{\rho}_{i,n_i} \eta_{i,n_i}}{\sqrt{\eta_{i,n_i}^2 + \sigma_{i,n_i}^2}} \quad (23)$$

where $\hat{\theta}_{i,q}$, $\hat{\rho}_{i,q}$ are the estimations of $\theta_{i,q}$ and $\rho_{i,q}$, respectively. The $\theta_{i,q}$ and $\rho_{i,q}$ are defined as follows:

$$\theta_{i,q} = \|W_{i,q}\|, \rho_{i,q} = \varepsilon_{i,q} \quad (24)$$

And $s_{i,q}$, $q = 1, \dots, n_i$ is a compensating signal designed as

$$\begin{aligned} \dot{s}_{i,1} = & -k_{i,1}s_{i,1} + (b_i + d_i)(\bar{\alpha}_{i,2} - \alpha_{i,1}) \\ & + (b_i + d_i)s_{i,2} - l_{i,1} \frac{(1 + \sigma_{i,1})s_{i,1}^2}{\sqrt{s_{i,1}^2 + \sigma_{i,1}^2}} \end{aligned} \quad (25)$$

$$\begin{aligned} \dot{s}_{i,q} = & -k_{i,q}s_{i,q} + \bar{\alpha}_{i,q+1} - \alpha_{i,q} - c_{i,q}s_{i,q-1} \\ & + s_{i,q+1} - l_{i,q} \frac{(1 + \sigma_{i,q})s_{i,q}^2}{\sqrt{s_{i,q}^2 + \sigma_{i,q}^2}} \quad q = 2, \dots, n_i - 1 \end{aligned} \quad (26)$$

$$\dot{s}_{i,n_i} = -k_{i,n_i}s_{i,n_i} - s_{i,n_i-1} - l_{i,n_i} \frac{\sigma_{i,n_i}s_{i,n_i}^2}{\sqrt{s_{i,n_i}^2 + \sigma_{i,n_i}^2}} \quad (27)$$

where $l_{i,j} > 0$ is a known constant, and $s_{i,j}(0) = 0$. Moreover,

$$c_{i,q} = \begin{cases} b_i + d_i, & q = 2 \\ 1, & 3 \leq q \leq n_i - 1 \end{cases} \quad (28)$$

Remark 3: In the control design, the virtual controllers in (16) and (17) include the term $\lambda_{i,q}\sigma_{i,q}\eta_{i,q}^{2p-1}$ ($q = 1, \dots, n_i - 1$), where $\frac{1}{2} < p < 1$. So, $2p - 1 > 0$ is ensured, i.e., the singularity problem can be avoided in the virtual controllers. Besides, in the traditional backstepping design, the control singularity problem may still arise from $\lambda_{i,q}\sigma_{i,q}\eta_{i,q}^{2p-1}$ when repeated differentiations of virtual controllers. To circumvent the singularity problem, the command filter backstepping control is employed without computing the first derivative of the virtual controllers. The singularity problem, as a result, is avoided.

To derive the main results, we perform the recursive control design as follows.

Step 1: The Lyapunov function candidate is chosen as

$$V_{i,1} = \frac{1}{2}\eta_{i,1}^2 + \frac{1}{2\gamma_{i,1}}\tilde{\theta}_{i,1}^2 + \frac{1}{2r_{i,1}}\tilde{\rho}_{i,1}^2 \quad (29)$$

where $\gamma_{i,1}$ and $r_{i,1}$ are known positive constants. $\eta_{i,j} = z_{i,j} - s_{i,j}$ denotes the compensated tracking error with $i = 1, 2, \dots, N, j = 1, 2, \dots, n_i$. And $\tilde{\theta}_{i,j} = \theta_{i,j} - \hat{\theta}_{i,j}$ and $\tilde{\rho}_{i,j} = \rho_{i,j} - \hat{\rho}_{i,j}$ denote the estimation errors. The time derivative of $V_{i,1}$ is

$$\begin{aligned} \dot{V}_{i,1} = & \eta_{i,1} \left[(b_i + d_i)(\eta_{i,2} + \alpha_{i,1}) + F_{i,1}(X_{i,1}) + k_{i,1}s_{i,1} \right. \\ & \left. + l_{i,1} \frac{s_{i,1}^2}{\sqrt{s_{i,1}^2 + \sigma_{i,1}^2}} + l_{i,1}\sigma_{i,1} \frac{s_{i,1}^2}{\sqrt{s_{i,1}^2 + \sigma_{i,1}^2}} \right] \\ & + \frac{1}{\gamma_{i,1}}\tilde{\theta}_{i,1}\dot{\hat{\theta}}_{i,1} + \frac{1}{r_{i,1}}\tilde{\rho}_{i,1}\dot{\hat{\rho}}_{i,1} \end{aligned} \quad (30)$$

where $F_{i,1}(X_{i,1}) = (b_i + d_i)f_{i,1} - b_i\dot{r} - \sum_{j=1}^N a_{ij}(x_{j,2} + f_{j,1}(x_{j,1}))$ is an unknown function with $X_{i,1} = (x_{i,1}, x_{j,2}, r, \dot{r})$. For any given constant $\varepsilon_{i,1}$, there exists an NN $W_{i,1}^T\psi_{i,1}(X_{i,1})$ such that $F_{i,1}(X_{i,1}) = W_{i,1}^T\psi_{i,1}(X_{i,1}) + \delta_{i,1}(X_{i,1})$, $|\delta_{i,1}| \leq \varepsilon_{i,1}$. By using Young's inequality, the following inequality holds $\eta_{i,1}F_{i,1}(X_{i,1}) \leq |\eta_{i,1}|(\theta_{i,1}\xi_{i,1} + \rho_{i,1}) + \sigma_{i,1}\theta_{i,1}$.

Then, the following inequality holds

$$\begin{aligned} \dot{V}_{i,1} \leq & \eta_{i,1} \left[(b_i + d_i)(\eta_{i,2} + \alpha_{i,1}) + k_{i,1}s_{i,1} \right. \\ & \left. + l_{i,1} \frac{(1 + \sigma_{i,j})s_{i,1}^2}{\sqrt{s_{i,1}^2 + \sigma_{i,1}^2}} \right] + \eta_{i,1}v_{i,j} \\ & + \frac{1}{\gamma_{i,1}}\tilde{\theta}_{i,1}(\gamma_{i,1}\eta_{i,1}\xi_{i,1} - \dot{\hat{\theta}}_{i,1}) + \sigma_{i,1}\theta_{i,1} \\ & + \frac{1}{r_{i,1}}\tilde{\rho}_{i,1} \left(r_{i,1} \frac{\eta_{i,1}^2}{\sqrt{\eta_{i,1}^2 + \sigma_{i,1}^2}} - \dot{\hat{\rho}}_{i,1} \right) + \sigma_{i,1}\rho_{i,1} \end{aligned} \quad (31)$$

Substituting (16), (19) and (20) into (31) produces

$$\begin{aligned} \dot{V}_{i,1} \leq & \eta_{i,1} [-k_{i,1}\eta_{i,1} - \lambda_{i,1}\sigma_{i,1}\eta_{i,1}^{2p-1} + (b_i + d_i)\eta_{i,2}] \\ & + \sigma_{i,1}(1 + \theta_{i,1} + \rho_{i,1}) + \frac{\sigma_{i,1}}{\gamma_{i,1}}\tilde{\theta}_{i,1}\hat{\theta}_{i,1}^{2p-1} \\ & + \frac{\sigma_{i,1}}{r_{i,1}}\tilde{\rho}_{i,1}\hat{\rho}_{i,1}^{2p-1} \end{aligned} \quad (32)$$

Step q: ($2 \leq q \leq n_i - 1$) The Lyapunov function candidate is chosen as

$$V_{i,q} = V_{i,q-1} + \frac{1}{2}\eta_{i,q}^2 + \frac{1}{2\gamma_{i,q}}\tilde{\theta}_{i,q}^2 + \frac{1}{2r_{i,q}}\tilde{\rho}_{i,q}^2 \quad (33)$$

where $\gamma_{i,q}$ and $r_{i,q}$ are positive constants.

Calculating the derivative of (33), we have

$$\begin{aligned} \dot{V}_{i,q} = & \dot{V}_{i,q-1} + \eta_{i,q} \left(\eta_{i,q+1} + F_{i,q}(X_{i,q}) \right. \\ & \left. + k_{i,q}s_{i,q} + \alpha_{i,q} + c_{i,q}s_{i,q-1} \right. \\ & \left. + l_{i,q} \frac{s_{i,q}^2}{\sqrt{s_{i,q}^2 + \sigma_{i,q}^2}} + l_{i,q}\sigma_{i,q} \frac{s_{i,q}^2}{\sqrt{s_{i,q}^2 + \sigma_{i,q}^2}} \right) \\ & - \frac{1}{\gamma_{i,q}}\tilde{\theta}_{i,q}\dot{\hat{\theta}}_{i,q} - \frac{1}{r_{i,q}}\tilde{\rho}_{i,q}\dot{\hat{\rho}}_{i,q} \end{aligned} \quad (34)$$

where $F_{i,q}(X_{i,q}) = f_{i,q} - \dot{\alpha}_{i,q}$ is an unknown nonlinear function with $X_{i,q} = (\bar{x}_{i,q}, \bar{x}_{j,q}, \hat{\theta}_{i,1}, \theta_{i,2}, \dots, \hat{\theta}_{i,q-1}, \hat{\rho}_{i,1}, \hat{\rho}_{i,2}, \dots, \hat{\rho}_{i,q-1})$. By using Lemma 4, one has $F_{i,q}(X_{i,q}) = W_{i,q}^T\psi_{i,q}(X_{i,q}) + \delta_{i,q}(X_{i,q})$, $|\delta_{i,q}| \leq \varepsilon_{i,q}$. Based on the Young's inequality, the following inequality can be obtained $\eta_{i,q}F_{i,q}(X_{i,q}) = \eta_{i,q}(W_{i,q}^T\psi_{i,q}(X_{i,q}) + \delta_{i,q}) \leq \theta_{i,q}\eta_{i,q}\xi_{i,q} + \sigma_{i,q}\theta_{i,q} + |\eta_{i,q}|\rho_{i,q}$. With the above-mentioned analysis, we could further get that

$$\begin{aligned} \dot{V}_{i,q} \leq & \dot{V}_{i,q-1} + \eta_{i,q} \left[\eta_{i,q+1} + k_{i,q}s_{i,q} + c_{i,q}s_{i,q-1} \right. \\ & \left. + \alpha_{i,q} + l_{i,q} \frac{(1 + \sigma_{i,q})s_{i,q}^2}{\sqrt{s_{i,q}^2 + \sigma_{i,q}^2}} \right] + \eta_{i,q}v_{i,q} \\ & + \frac{1}{\gamma_{i,q}}\tilde{\theta}_{i,q}(\gamma_{i,q}\eta_{i,q}\xi_{i,q} - \dot{\hat{\theta}}_{i,q}) + \sigma_{i,q}\theta_{i,q} \\ & + \frac{1}{r_{i,q}}\tilde{\rho}_{i,q} \left(r_{i,q} \frac{\eta_{i,q}^2}{\sqrt{\eta_{i,q}^2 + \sigma_{i,q}^2}} - \dot{\hat{\rho}}_{i,q} \right) + \sigma_{i,q}\rho_{i,q} \end{aligned} \quad (35)$$

$s_{i,n_i} \dot{s}_{i,n_i}$). According to the (25), (26) and (27), the following equality holds:

$$\begin{aligned} \dot{V} = & \sum_{i=1}^N \left[- \sum_{j=1}^{n_i} k_{i,j} s_{i,j}^2 + \sum_{j=1}^{n_i-1} c_{i,q} (\bar{\alpha}_{i,j} - \alpha_{i,j-1}) |s_{i,j}| \right. \\ & \left. - \sum_{j=1}^{n_i-1} \frac{l_{i,j} s_{i,j}^2}{\sqrt{s_{i,j}^2 + \sigma_{i,j}^2}} - \sum_{j=1}^{n_i} \frac{l_{i,j} \sigma_{i,j} s_{i,j}^2}{\sqrt{s_{i,j}^2 + \sigma_{i,j}^2}} \right] \end{aligned} \quad (45)$$

Using the fact that $|\bar{\alpha}_{i,j} - \alpha_{i,j-1}| \leq l_{i,j}$ in [25], we can obtain the following inequality:

$$\begin{aligned} \dot{V} \leq & \sum_{i=1}^N \left(- \sum_{j=1}^{n_i} k_{i,j} s_{i,j}^2 - \sum_{j=1}^{n_i} l_{i,j} \sigma_{i,j} |s_{i,j}| + \sum_{j=1}^{n_i} 2l_{i,j} \sigma_{i,j} \right) \\ \leq & -k_0 \bar{V}^{\frac{1}{2}} + m_0 \end{aligned} \quad (46)$$

where $k_0 = 2 \min\{l_{i,j} \sigma_{i,j}\}$, $m_0 = \sum_{i=1}^N \sum_{j=1}^{n_i} 2l_{i,j} \sigma_{i,j}$ and $l_{i,j}$ is chosen such that $l_{i,j} - c_{i,q} l_{i,j} > 0$. Through a similar analysis, the compensating signal $s_{i,j}$ will converge to the compact set $\Omega_0 = \{s_{i,j} | s_{i,j} \leq \frac{m_0}{(1-\varrho_1)l_0}\}$ in finite-time, and the convergence time T_2 is calculated as $\frac{1}{\lambda} \ln \frac{\varrho_1 l_0}{2\lambda(\frac{m_0}{(1-\varrho_1)l_0} - \bar{V}(0)^{\frac{1}{2}}) + \varrho_1 l_0}$. By using the fact of $z_{i,j} = \eta_{i,j} + s_{i,j}$, one can conclude that $z_{i,j}$ is finite-time stable with $T \leq T_1 + T_2$. Note that the virtual control signal $\alpha_{i,j}$ is the function of bounded signals $\eta_{i,j}$ and $z_{i,j}$, it is straightforward to show that the virtual control signals $\alpha_{i,j}$ is bounded. This, together with the relationship $z_{i,q} = x_{i,q} - \bar{\alpha}_{i,q}$, implies that the boundedness of $x_{i,q}$ can be obtained. In addition, the tracking error $z_{i,1}$ converges to the set $\Omega_1 = \{z_{i,1} | (\frac{b}{(1-\varrho)w})^{\frac{1}{2p}} + \frac{m_0}{(1-\varrho_1)l_0}\}$ within finite-time $T \leq T_1 + T_2$.

In particular, we can prove that the consensus tracking errors can reach zero as time approaches to infinity. The proof is given below.

Integrating (44) from 0 to t , then (44) can become the following one:

$$\begin{aligned} V(t) - V(0) \leq & \sum_{i=1}^N \sum_{j=1}^{n_i} \left[- \int_0^t \lambda_{i,j} \eta_{i,j}^{2p} d\tau + \int_0^t \delta_{i,2}^2 e^{-2\beta\tau} d\tau \right. \\ & + \int_0^t \left(\frac{\theta_{i,j}^2}{2} + \frac{\rho_{i,j}^2}{2} + \frac{d_0}{\gamma_{i,j}} \theta_{i,j}^{2p} + \frac{d_0}{r_{i,j}} \rho_{i,j}^{2p} \right) \\ & \left. \times \sigma_{i,j} d\tau + \int_0^t \sigma_{i,j} (1 + \theta_{i,j} + \rho_{i,j}) d\tau \right] \end{aligned} \quad (47)$$

The above mentioned formulas can be used to derive the following expression:

$$\begin{aligned} & \sum_{i=1}^N \sum_{j=1}^{n_i} \int_0^t \lambda_{i,j} \eta_{i,j}^{2p}(\tau) d\tau \\ & \leq \sum_{i=1}^N \sum_{j=1}^{n_i} \left[V(0) + \frac{1}{\lambda} (1 + \theta_{i,j} + \rho_{i,j}) + \frac{\delta_{i,2}}{2\beta} \right. \\ & \quad \left. + \frac{1}{\lambda} \left(\frac{\theta_{i,j}^2}{2} + \frac{\rho_{i,j}^2}{2} + \frac{d_0}{\gamma_{i,j}} \theta_{i,j}^{2p} + \frac{d_0}{r_{i,j}} \rho_{i,j}^{2p} \right) \right] < \infty \end{aligned} \quad (48)$$

Hence, by using Lemma 2, we can infer that $\lim_{t \rightarrow +\infty} \eta_{i,j} = 0$. Next, the convergence of compensating signal $s_{i,j}$ will be

proved. By integrating the left and right sides of (46) on $[0, t)$, we obtain

$$\begin{aligned} & \bar{V}(t) - \bar{V}(0) \\ & \leq \sum_{i=1}^N \sum_{j=1}^{n_i} \left[-k_0 \int_0^t s_{i,j}^2(\tau) d\tau + 2 \frac{l_{i,j}}{\lambda} (1 - e^{-\lambda t}) \right] \end{aligned} \quad (49)$$

From (49), the following inequality holds

$$\begin{aligned} & \sum_{i=1}^N \sum_{j=1}^{n_i} k_0 \int_0^t s_{i,j}^2(\tau) d\tau \leq \bar{V}(0) + 2 \frac{l_{i,j}}{\lambda} (1 - e^{-\lambda t}) \\ & \leq \bar{V}(0) + 2 \frac{l_{i,j}}{\lambda} < \infty \end{aligned} \quad (50)$$

In consideration of Lemma 2, we can verify that $\lim_{t \rightarrow +\infty} s_{i,j} = 0$. This, together with the relation $\eta_{i,j} = z_{i,j} - s_{i,j}$, and the asymptotically convergence of $\eta_{i,j}$ and $s_{i,j}$, we can infer that $\lim_{t \rightarrow +\infty} z_{i,j} = 0$. In the following, we will prove the asymptotic tracking. By defining $z_1 = [z_{i,1}, z_{i,2}, \dots, z_{N,1}]^T$, we have

$$z_1 = (\mathcal{L} + \mathcal{B})\delta \quad (51)$$

where $\delta = \underline{y} - \underline{y}_r$ with $\underline{y} = [y_1, \dots, y_N]^T \in R^N$ and $\underline{y}_r = [r(t), \dots, r(t)]^T \in R^N$ denotes the actual tracking error between each subsystem's output and y_r . $\mathcal{B} = \text{diag}\{b_i\}$. Due to $\mathcal{L} + \mathcal{B}$ being positive definite, it follows that asymptotic consensus tracking of all the N subsystems' outputs to a desired trajectory $r(t)$ is also achieved, i.e. $\lim_{t \rightarrow \infty} \delta_i(t) = 0$ for $i = 1, \dots, N$, which means that the asymptotic tracking is achieved.

Then, we prove that the Zeno behavior is avoided. By noting that: $e_{is} = u_{is} - u_i(t)$, $t_i \in [t_{i,k}, t_{i,k+1})$. Then, we have

$$\frac{d}{dt} |e_{is}| = \frac{d}{dt} (e_{is} * e_{is})^{\frac{1}{2}} = \text{sign}(e_{is}) \dot{e}_{is} \leq |\dot{u}_{is}| \quad (52)$$

From the above analysis, one has $|\dot{u}_{is}|$ is bounded since $|\dot{u}_{is}|$ is a function of bounded signals. Thus, the following inequality holds: $|\dot{u}_{is}| \leq v_i$, where v_i is a positive constant. By using the fact $e_{is}(t_{i,k}) = 0$ and $\frac{d}{dt} |e_{is}| \leq v_i$, one has $|e_{is}| \leq v_i(t - t_{i,k})$, $t \in [t_{i,k}, t_{i,k+1})$. Recalling the ETC mechanism, the following equation is derived

$$|e_{is}(t_{i,k+1})| = \delta_{i,1} |u_{is}(t_{i,k+1})| + \delta_{i,2} e^{-\beta t_{i,k+1}}, t = t_{i,k+1} \quad (53)$$

Therefore, the following inequality holds

$$\delta_{i,2} e^{-\beta t_{i,k+1}} \leq |e_{is}(t_{i,k+1})| \leq v_i T_{i,k}, \text{ with } T_{i,k} = t_{i,k+1} - t_{i,k} \quad (54)$$

It's easy to derive $T_{i,k} > 0$ for any finite interval. The proof will be accomplished by a contradiction. Suppose that $t_{i,\infty} = \lim_{k \rightarrow \infty} t_{i,k} < \infty$, which means that $\lim_{k \rightarrow \infty} T_{i,k} = 0$. By combining (54), it follows that $k \rightarrow \infty$, $0 < \delta_{i,2} e^{-\beta t_{i,\infty}} \leq v_i 0 = 0$. Therefore, Zeno behavior is excluded.

Remark 5: Different from [25], the frequency of controller updates in our paper can be effectively reduced by designing a time-varying threshold event-triggered mechanism. In addition, it is worth mentioning that the majority of finite-time works cannot reach asymptotic convergence, which is undesirable in

some systems requiring higher precision. In this paper, a new finite-time ETC method is designed for the MASs, where the tracking error can achieve asymptotically convergence.

V. SIMULATION STUDY

Consider the following nonlinear MASs, which is composed of one leader and four followers with the topology graph described by Fig. 1. Each follower is described by

$$\begin{aligned} \dot{x}_{i,1} &= x_{i,2} + \sin(x_{i,1}) \\ \dot{x}_{i,2} &= -22 \sin(x_{i,1}) - 2x_{i,1} + 10u_i, i = 1, 2, 3, 4 \end{aligned} \quad (55)$$

The trajectory of leader is given as $r(t) = 0.95 \sin(t)$.

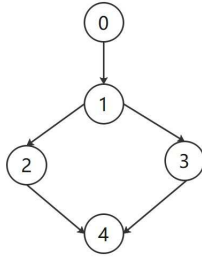
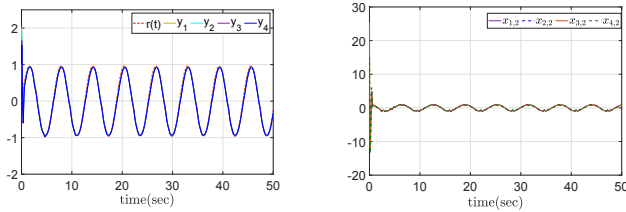


Fig. 1. Directed interaction topology for the 4 followers and 1 leader.

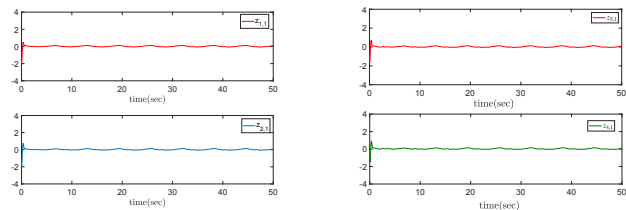
The design parameters are set as $[k_{1,1}, k_{1,2}, k_{2,1}, k_{2,2}, k_{3,1}, k_{3,2}, k_{4,1}, k_{4,2}] = [200, 200, 200, 200, 200, 200, 200, 200]$. $[l_{1,1}, l_{1,2}, l_{2,1}, l_{2,2}, l_{3,1}, l_{3,2}, l_{4,1}, l_{4,2}] = [1, 3, 2, 4, 1, 3, 2, 4]$. $[\lambda_{1,1}, \lambda_{1,2}, \lambda_{2,1}, \lambda_{2,2}, \lambda_{3,1}, \lambda_{3,2}, \lambda_{4,1}, \lambda_{4,2}] = [0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1]$. The initial values of $[x_{1,1}, x_{1,2}, x_{2,1}, x_{2,2}, x_{3,1}, x_{3,2}, x_{4,1}, x_{4,2}]$ are set as $[1, 1, 1, 1, 2, 2, 2, 2]$. Moreover, the parameters of $\epsilon_{i,q}$ are designed as $\epsilon_{i,2} = 0.1, i = 1, 2, 3, 4$.

The simulation results are shown in Figs. 2-8. As shown in Fig. 2(a), all follower outputs can track leader $r(t)$ in finite-time under our developed control algorithm. Fig. 2(b) reveals that the



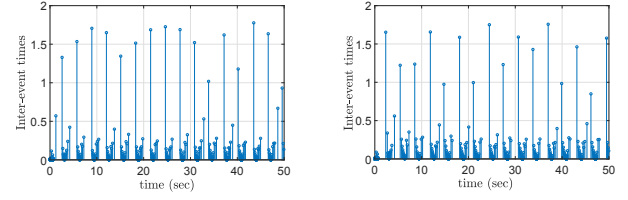
(a) The trajectories of $r(t)$ and y_i , $i = 1, 2, 3, 4$. (b) The trajectories of $x_{1,2}, x_{2,2}, x_{3,2}$ and $x_{4,2}$.

Fig. 2. The trajectories of $r(t)$ and $x_{i,j}, i = 1, 2, 3, 4, j = 1, 2$.



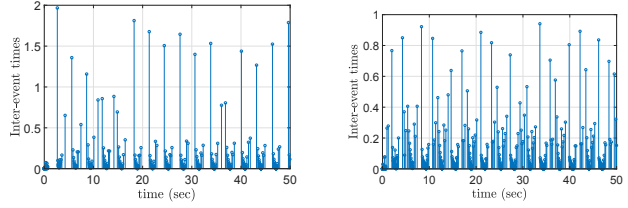
(a) Tracking errors of $z_{1,1}$ and $z_{2,1}$. (b) Tracking errors of $z_{3,1}$ and $z_{4,1}$.

Fig. 3. Tracking errors of $z_{i,1}, i = 1, 2, 3, 4$.



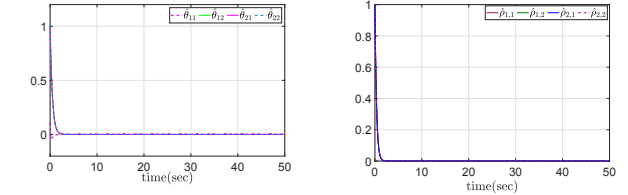
(a) Inter-event times of agent 1. (b) Inter-event times of agent 2.

Fig. 4. Inter-event times of agents 1 and 2.



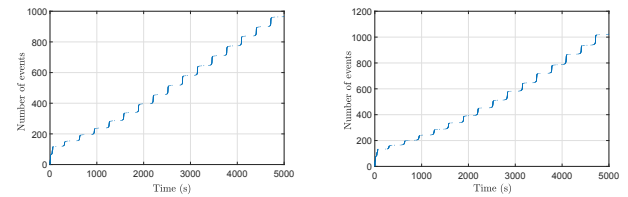
(a) Inter-event times of agent 3. (b) Inter-event times of agent 4.

Fig. 5. Inter-event times of agents 3 and 4.



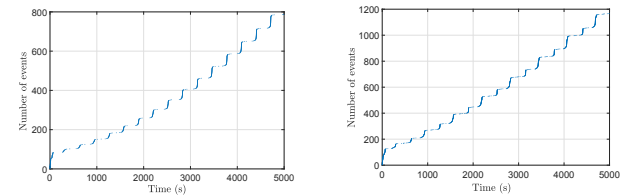
(a) Adaptive parameters of $\hat{\theta}_{i,j}$. (b) Adaptive parameters of $\hat{\rho}_{i,j}$.

Fig. 6. Adaptive parameters of $\hat{\theta}_{i,j}$ and $\hat{\rho}_{i,j}$.



(a) Cumulative number of events for agent 1. (b) Cumulative number of events for agent 2.

Fig. 7. Cumulative number of events for agent 1, 2.



(a) Cumulative number of events for agent 3. (b) Cumulative number of events for agent 4.

Fig. 8. Cumulative number of events for agent 3, 4.

states $x_{1,2}$, $x_{2,2}$, $x_{3,2}$ and $x_{4,2}$ are bounded. Fig. 3 illustrates the tracking errors of $z_{i,1}$, $i = 1, 2, 3, 4$. From the Fig. 3, the error converges to zero which means the followers can track the leader accurately. Figs. 4-5 show that the time of the released interval of triggered events. In addition, the triggered times of event triggering conditions of different agent are different, indicating that the event-triggered mechanism is distributed. The boundedness of the adaptive laws can be observed in Fig 6. Figs 7-8 show that the ETC above can effectively reduce the amount of data transmissions. The simulation results highlight that the proposed control algorithm could achieve the control objective.

VI. CONCLUSIONS

This paper has presented a finite-time asymptotic tracking control protocol for nonlinear MASs under a directed graph. In the proposed scheme, a novel event-triggered mechanism with a time-varying threshold is designed to reduce the frequency of controller updates and save network resources. Besides, both the singularity and explosion of complexity problems have been avoided by employing the command filtered technique. Then, a rigorous theoretical analysis is performed to demonstrate that the signals in the resulting systems are bounded in finite-time. In addition, the asymptotic tracking control can be obtained by using the same adaptive framework.

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